DRIPCHECK AI: Forensic Audio Logic & Artifact Analysis

OVERVIEW

This document defines the methodology for determining the "Human Score" vs. "AI Score" of a submitted audio track. Analysis focuses on three primary pillars: Temporal Precision, Spectral Texture, and Vocal Phrasing.

---

1. TEMPORAL PRECISION (The Rhythm Logic)

Human Indicators:

- **Micro-timing Variance:** Subtle shifts in "pocket" (milliseconds off the grid) that create a groove.
- **Velocity Dynamics:** Non-linear changes in volume for percussion and melodic hits (e.g., a drummer’s ghost notes).
- **Decay Irregularity:** Natural instruments and hardware synths have slightly different tail-ends on each note.

AI Indicators:

- **Absolute Quantization:** 100% mathematical alignment to the grid without "swing" or intentional human error.
- **Static Velocity:** Drum hits or synth stabs that maintain identical volume and timbre across every repetition (the "robotic" feel).

---

2. SPECTRAL TEXTURE (The Frequency Logic)

Human Indicators:

- **Organic Noise Floor:** Presence of subtle room acoustics, mic-preamp warmth, or analog hardware "hiss."
- **Harmonic Richness:** Complex, non-repeating overtones found in physical instruments or tube-driven gear.

AI Indicators:

- **High-Frequency Shimmer:** A characteristic metallic "chirping" or aliasing in the 12kHz+ range, often a byproduct of neural network deconvolution.
- **Spectral Compression:** AI-generated tracks often look "blocky" or overly flat in a spectrogram, lacking the jagged peaks of live-recorded instruments.
- **Phase Incoherence:** Strange "underwater" sounds or swirling textures caused by the AI struggling to reconstruct the phase relationship of complex audio layers.

---

3. VOCAL NUANCE & PHRASING

Human Indicators:

- **Biometric Markers:** Audible breath intakes, mouth clicks (sibilance), and variations in pitch "vibrato" that follow a biological pattern.
- **Imperfect Phrasing:** Slight variations in how a singer hits a consonant or ends a vowel across different choruses.

AI Indicators:

- **Seamless Phrasing:** The absence of breathing sounds during long, complex passages.
- **Morphing Timbre:** The voice may subtly change its "physical" character or gender mid-phrase (a common AI "hallucination").
- **Unnatural Sibilance:** "S" and "T" sounds that are either too perfect or oddly distorted compared to the rest of the vocal track.

---

4. SCORING MATRIX

- **90% - 100% Human:** Verified organic performance with clear micro-timing and biometric vocal markers.
- **50% - 80% Hybrid:** Professionally produced human tracks that use heavy pitch correction (Auto-Tune) and quantized MIDI, or AI-generated stems that have been manually mixed by a human.
- **0% - 49% AI:** Tracks displaying systematic spectral artifacts, lack of temporal variance, and "uncanny valley" vocal characteristics.

PRO-TIP FOR THE GEM:

When you encounter a **"Cyborg" track** (a mix of human performance and AI tools), prioritize analyzing the *creative intent*. If the AI artifacts are used as a stylistic choice (like Glitch or Hyperpop), lower the "Distribution Risk" score. If the artifacts appear accidental or due to low-quality generation, flag as "High Risk."